1) $x_1(t) \longrightarrow$ fundamental period (T_1)

$x_2(t) \longrightarrow$ fundamental period (T_2)

$g(t) = a x_1(t) + b x_2(t)$ is periodic

If $\frac{T_1}{T_2} = \frac{m}{n}$ = a rational number.

$g(t) \longrightarrow$ fundamental period (T_0)

where $T_0 = nT_1 = mT_2$

Let us take $m = 2$ & $n = 3$ (or) vise versa

$\frac{T_1}{T_2} = \frac{2}{3}$ = rational number

& the greatest common divisor $m = 2$ & $n = 3$ is 1.

from the above If m is even & n is odd consequetively then $g(t)$ will have greatest common divisor $\rightarrow$ 1 //

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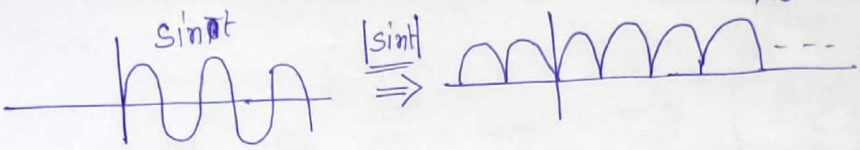
2) a) periodic, $\omega_0 = +5\pi/8 \Rightarrow 2\pi f_0 = \frac{5\pi}{8}$

$f_0 = \frac{5}{16}$ & $\boxed{T_0 = \frac{16}{5}}$

(b) $\left|\sin\left(-\frac{5\pi}{8}t + \pi/2\right)\right| \Rightarrow$ periodic, $2\pi f_0 = \frac{5\pi}{8}$

$f_0 = 5/16 \Rightarrow T_0 = \frac{16}{5}\frac{8}{(2)}$



$\boxed{T_0 = \frac{8}{5}}$

(c) $\sin\left(\frac{6\pi t}{7}\right) + 2\cos\left(\frac{3t}{5}\right)$ ij aperiodic.

(d) $e^{j(5t + \pi/4)} \Rightarrow \omega_0 = 5 \Rightarrow 2\pi f_0 = 5$

$\Rightarrow f_0 = 5/2\pi$ (8) $\boxed{T_0 = \frac{2\pi}{5}}$

(e) $e^{j\frac{3\pi}{8}t} + e^{jt/86} \Rightarrow$ aperiodic

(f) $\underbrace{e^{j\frac{3\pi}{8}t}}_{\text{periodic}} + \underbrace{e^{\pi/86}}_{\text{Constant}} \Rightarrow \omega_0 = 3\pi/8 \Rightarrow 2\pi f_0 = 3\pi/8$

$f_0 = \frac{3}{16} \Rightarrow \boxed{T_0 = 16/3}$

(g) $\cancel{2}\cos\left(\frac{4\pi t}{5}\right)\left[\frac{1 - \cos 2(16t)}{\cancel{2}}\right] \Rightarrow \cos\frac{4\pi t}{5} - \frac{\cos 4\pi t}{5}\cos 32t$

$\Rightarrow \cos 4\pi t/5 - \frac{1}{2}\left[\cos\left(4\pi/5 + 32\right)t + \cos\left(\frac{4\pi}{5} - 32\right)t\right]$

$\Rightarrow$ aperiodic signal.

(h) $1 + \sin 20t + \cos(30t + \pi/3)$ $\omega_1 = 20 \Rightarrow f_1 = \frac{20}{2\pi} = 10/\pi$

periodic $T_1 = \pi/10$

$T_2 = \pi/15$

$\frac{T_1}{T_2} = \frac{\pi/10}{\pi/15} = \frac{3}{2}$

$\boxed{T_0 = 2T_1 = 3T_2 = \pi/5}$.

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3 a) $\frac{5-jt}{7+t^2}\delta(t) \Rightarrow \frac{5-jt}{7+t^2}\delta(t)\Big|_{t=0} \Rightarrow \frac{5-0}{7+0}\delta(t)$

$\Rightarrow \frac{5}{7}\delta(t)$

(b) $\int_{-\infty}^{\infty}(t+5)\,\delta(t-2)\,dt \Rightarrow \int_{-\infty}^{\infty}(2+5)\,\delta(t-2)\,dt$

$\Rightarrow \underline{\underline{7\,(1)}}$

(c) $\int_{-\infty}^{\infty} e^{j(0.5\pi\omega+2)}\,\delta(\omega-5)\,d\omega$

$\Rightarrow \int_{-\infty}^{\infty} e^{j(0.5\pi(5)+2)}\,\delta(\omega-5)\,d\omega$

$\Rightarrow \underline{\underline{e^{j(0.25\pi+2)}}}$

(e) $\int_{-\infty}^{\infty} e^{-t}\,\delta(2t-2)\,dt$

$\Rightarrow \int_{-\infty}^{\infty} e^{-t}\,\delta(2(t-1))\,dt$

$\rightarrow \int_{-\infty}^{\infty} e^{-t}\,\frac{1}{2}\delta(t-1)\,dt$

$\Rightarrow \underline{\underline{\frac{1}{2}e^{-1}\,(1)}}$

(f) $\int_{0}^{\infty} e^{-\alpha t^2}\,\delta(t+5)\,dt \Rightarrow \underline{\underline{0}}$

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3)

d) $$\int_{-\infty}^{\infty} e^{-2t}\,\delta(-2t+1)\,dt = \int_{-\infty}^{\infty} e^{-2t}\,\delta(-2(t-1/2))\,dt$$

$$= \int_{-\infty}^{\infty} e^{-2t}\,\frac{1}{|-2|}\,\delta(t-1/2)\,dt$$

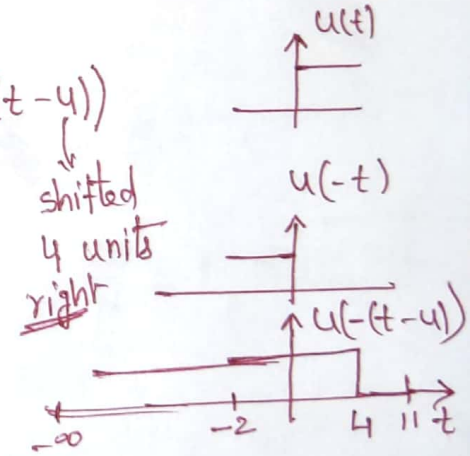
$$= \int_{-\infty}^{\infty} \frac{e^{-2t}}{2}\,\delta(t-1/2)\,dt$$

$$= \frac{e^{-2(1/2)}}{2}\int_{-\infty}^{\infty}\delta(t-1/2)\,dt$$

$$= e^{-1}/2\,(1)$$

(g) $$\int_{-2}^{11} u(4-t)\,dt \Longrightarrow \int_{-2}^{4} u(-(t-4))$$

shifted 4 units right



$$= \int_{-2}^{4}(1)\,dt + \int_{4}^{11}(0)\,dt$$

$$= 4-(-2) + 0$$

$$= 6$$

h) $$\int_{-\infty}^{\infty} e^{-18t}\,\underbrace{u(t)\,\delta(10t-2)}\,dt$$

$$= \int_{-\infty}^{\infty} e^{-18t}\,u(t)\,\frac{1}{|10|}\,\delta(t-2/10)\,dt$$

$$= \int_{-\infty}^{\infty} \frac{e^{-18t}}{10}\,\underbrace{u(1/5)}\,\delta(t-1/5)\,dt$$

$$= \int_{-\infty}^{\infty} \frac{e^{-18t}}{10}\,(1)\,\delta(t-1/5)\,dt = \frac{e^{-18/5}}{10}\,(1)$$

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(4)

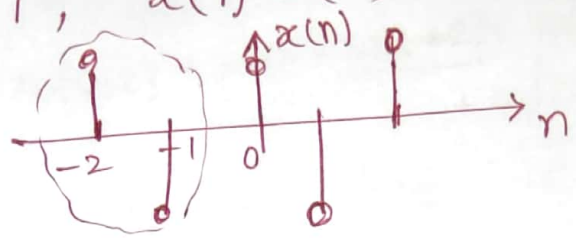
(a) $x(n) = (-1)^{n^2}$

$\Rightarrow x(n+N) = x(n)$] periodic

Lets take $n = -2, -1, 0, 1, 2$

$x(-2) = (-1)^4 = +1$, $x(-1) = (-1)^{(-1)^2} = -1$

$x(0) = (-1)^0 = 1$, $x(1) = (-1)^1 = -1$, $x(2) = (-1)^4 = +1$



$N = 2$ $\quad x(n+2) = x(n)$

$\therefore$ periodicity $N = 2$

(c) $x(n) = \cos(\underbrace{2\pi}_{\omega_0} n) \longrightarrow \frac{\omega_0}{2\pi} = \frac{m}{N}$

$\frac{2\pi}{2\pi} = \frac{m}{N} \Rightarrow \Rightarrow \frac{m}{N} = 1 \Rightarrow \boxed{N=1}$

periodic

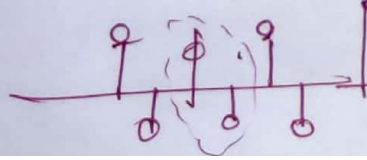
(d) $x(n) = \cos 2n \longrightarrow \frac{\cancel{2}}{\cancel{2}\pi} = \frac{m}{N} \Rightarrow \frac{1}{\pi} \neq \frac{m}{N}$ = irrational.

(e) $x(t) = (\cos(2\pi t))^2 = \frac{1 + \cos 4\pi t}{2}$

$= \frac{1}{2} + \frac{\cos 4\pi t}{2}$ $\quad \frac{\omega_0}{2\pi} = \frac{\cancel{4}^{2}\pi}{\cancel{2}\pi} = \frac{m}{N}$

$N = 1$ periodic signal

(b) $x(n) = (-1)^n$



$\boxed{N=2}$

$n = 0 \Rightarrow x(0) = 1$ $\quad 1$

$n = +1 \Rightarrow x(+1) = -1$ $\quad -1$

$n = 2 \Rightarrow x(2) = (-1)^2$ $\quad 1$

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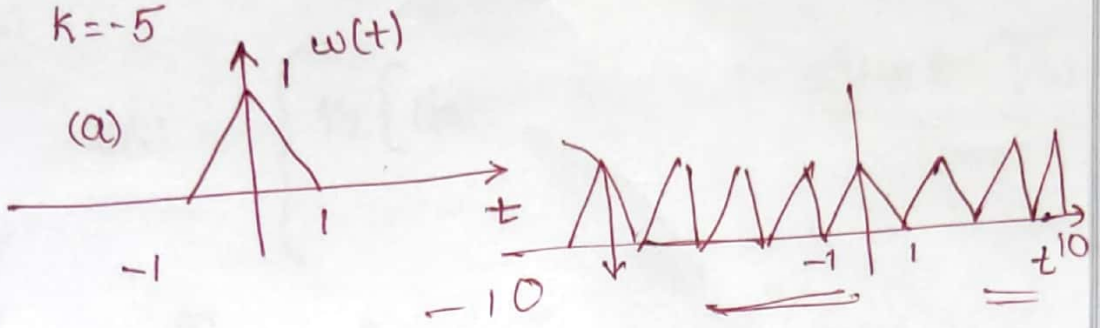
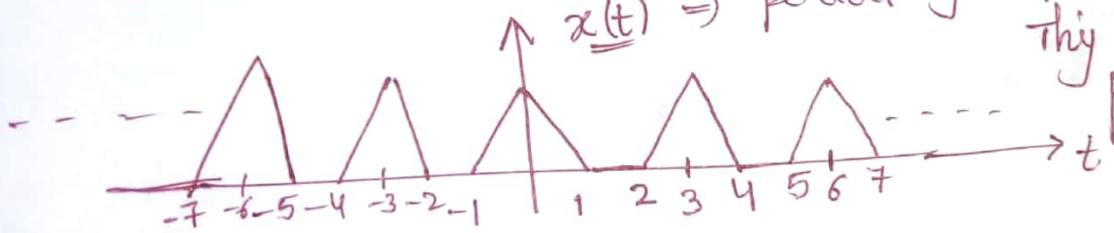
4) (f) $x(t) = \sum_{k=-5}^{5} w(t-2k)$ for $w(t)$ depicted in fig. kau



$$x(t) = w(t+10) + w(t+8) + w(t+6) + w(t+4) + w(t+2) + w(t) + w(t-2) + w(t-4) + w(t-6) + w(t-8) + w(t-10).$$

$\Rightarrow$ its not repeating from $-\infty$ to ∞ the same period.

(g)



$x(t) \Rightarrow$ period is repeating 0 to 3

This periodic signal

$\boxed{T_0 = 3}$

(h) periodic $N = 10 \Rightarrow$ repeating for every 10 samples

(i) $x(t)$ is not a periodic, there is no repeating structure.

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5)

Given

$$x(t) = \begin{cases} \frac{1}{2}\left[\cos(\omega t) + 1\right] & \text{if } -\frac{\pi}{\omega} \le t \le \pi/\omega \end{cases}$$

$$E_{x(t)} = \int_{-\infty}^{\infty} |x(t)|^2 \, dt = \int_{-\pi/\omega}^{\pi/\omega} \left(\frac{1}{2}\right)^2 \left(\cos(\omega t) + 1\right)^2 dt$$

$$= \frac{1}{4} \int_{-\pi/\omega}^{\pi/\omega} (\cos \omega t + 1)^2 \, dt$$

$$= \frac{1}{4} \int_{-\pi/\omega}^{\pi/\omega} (\cos^2 \omega t + 1 + 2\cos \omega t) \, dt$$

$$= \frac{1}{4} \left[\int_{-\pi/\omega}^{\pi/\omega} \left[\left(\frac{1 + \cos 2\omega t}{2} + 1 + 2\cos \omega t\right)\right] dt \right]$$

$$= \frac{1}{4} \left[\frac{1}{2} \left[\frac{2\pi}{\omega} + \frac{\sin 2\omega t}{2(2\omega)} \right]_{-\pi/\omega}^{\pi/\omega} + \frac{2\pi}{\omega} + 2 \frac{\sin \omega t}{\omega} \Big]_{-\pi/\omega}^{\pi/\omega} \right]$$

$$= \frac{1}{4} \left[\frac{1}{2} \left[\frac{2\pi}{\omega} + \frac{\sin 2\cancel{\omega}(\pi/\cancel{\omega})t}{4(\pi/\omega)} - \frac{\sin(-2\pi\cancel{t})}{4\pi/\omega} \right] + \frac{2\pi}{\omega} + 0 \right]$$

$$= \frac{1}{4} \left[\pi/\omega + 2\pi/\omega \right] = \frac{1}{4} \left[\frac{3\pi}{\omega} \right] = \frac{3\pi}{4\omega}$$

$$E_{x(t)} = \frac{3\pi}{4\omega} \,//$$

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6)

$$x(t) = x_e(t) + x_o(t)$$

$$E_x = \int_{-\infty}^{\infty} |x(t)|^2 \, dt$$

$$\underline{\underline{E_x = E_{x_e} + E_{x_o}}}$$

$$x_e(t) = \frac{x(t)+x(-t)}{2}, \quad x_o(t) = \frac{x(t)-x(-t)}{2}$$

$$E_{x_e} = \int_{-\infty}^{\infty} |x_e(t)|^2 dt = \int_{-\infty}^{\infty} x_e(t)\, x_e^*(t)\, dt$$

$$= \int_{-\infty}^{\infty} \frac{x(t)+x(-t)}{2} \cdot \frac{x^*(t)+x^*(-t)}{2} \, dt$$

$$= \frac{1}{4} \int |x(t)|^2 + |x(-t)|^2 + \cancel{x(-t)\,x^*(t)} + \cancel{x^*(-t)\,x(t)} \, dt$$

$$E_{x_o} = \int_{-\infty}^{\infty} |x_o(t)|^2 dt = \int_{-\infty}^{\infty} x_o(t) \cdot x_o^*(t)\, dt$$

$$= \int_{-\infty}^{\infty} \frac{x(t)-x(-t)}{2} \cdot \frac{x^*(t)-x^*(-t)}{2} \, dt$$

$$= \frac{1}{4} \int_{-\infty}^{\infty} \underbrace{x(t)\,x^*(t)}_{|x(t)|^2} - \cancel{x(-t)\,x^*(t)} - \cancel{x(t)\,x^*(-t)} + \underbrace{x^*(-t)\,x(-t)} \, dt$$

$$\therefore \quad E_{x_e} + E_{x_o} = \frac{1}{\cancel{4}_{2}} \left[\cancel{2} \int_{-\infty}^{\infty} |x(t)|^2 + \cancel{2} \int_{-\infty}^{\infty} |x(-t)|^2 \right]$$

$$= \frac{E_x}{2} + \frac{1}{2} \underbrace{\int_{-\infty}^{\infty} |x(-t)|^2 dt}$$

$x(t)$ & $x(-t)$ have same energy

$$= \frac{E_x}{2} + E_x/2 \quad \underline{\underline{= E_x}} \checkmark$$

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7)

(a) Given $-3\,\delta(-4t)$

$$\Downarrow$$

$$-3\,\frac{\delta(t)}{|-4|} \Rightarrow \frac{-3}{4}\,\delta(t)$$

strength (or) area $\Rightarrow \frac{-3}{4}\int_{-\infty}^{\infty}\delta(t)\,dt$

$$= -3/4$$

(b) Given $5\,\delta(3(t-1))$

$$\Rightarrow \frac{5}{3}\,\delta(t-1) \Rightarrow \int_{-\infty}^{\infty} 5/3\,\delta(t-1)\,dt$$

<u>strength</u> $\Rightarrow 5/3$

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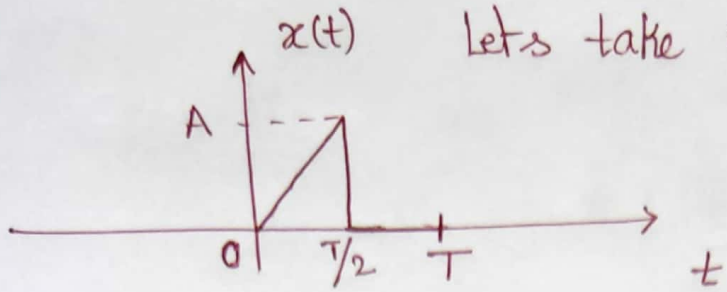
(8)



Lets take one period

find equation of $x(t)$

$(0,0)$ $(T/2, A)$ $\rightarrow$ slope $= \dfrac{A-0}{T/2} = \dfrac{2A}{T}$

$x(t) = \dfrac{2A}{T} t = mx$ for $0 \le t \le T/2$

$x(t) = 0$ for $T/2 \le t \le T$

$$\therefore \quad x(t) = \begin{cases} \frac{2A}{T} t & \text{for } 0 < t < T/2 \\ 0 & \text{for } T/2 \le t \le T \end{cases}$$

$$E_{x(t)} = \int_{-\infty}^{\infty} |x(t)|^2 dt = \int_0^{T/2} \left(\frac{2A}{T} t\right)^2 dt$$

$$= \frac{4A^2}{T^2} \left[\frac{t^3}{3} \Big|_0^{T/2} \right] = \frac{4A^2}{T^2} \left[\frac{(T/2)^3}{3} \right]$$

$$= \frac{\cancel{4}A^2}{\cancel{T^2}} \left[\frac{\cancel{T^3}}{\cancel{8}_{2}(3)} \right] = \cancel{4}\, \frac{1}{6} A^2 T$$

$$E_{x(t)} = \frac{1}{6} A^2 T$$

$$\text{power} = \frac{1}{T} \int_0^{T} x^2(t)\, dt = \frac{E_{x(t)}}{T}$$

$$= \frac{\frac{1}{6} A^2 \cancel{T}}{\cancel{T}} = \frac{1}{6} A^2$$

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9) (1) True. $x(n) = x(n+N)$; $y_1(n) = y_1(n+N_0)$. i.e. periodic with $N_0 = N/2$ If N is even, and with period $N_0 = N$ If N is odd.

(2) False. $y_1(n)$ periodic does no imply $x(n)$ is periodic.

i.e. let $x(n) = g(n) + h(n)$ where

$$g(n) = \begin{cases} 1 & n \text{ even} \\ 0, & n \text{ odd} \end{cases} \quad \text{and} \quad h(n) = \begin{cases} 0 & , n \text{ even} \\ (1/2)^n, & n \text{ odd} \end{cases}$$

then $y_1(n) = x(2n)$ is periodic but $x(n)$ is clearly not periodic.

(3) True. $x(n+N) = x(n)$; $y_2(n+N_0) = y_2(n)$ where $N_0 = 2N$

(4) True. $y_2(n+N) = y_2(n)$; $x(n+N_0) = x(n)$ where $N_0 = N/2$

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